

Substituting equations (11a)-(11c) into the DD decomposition theorem expresses the probability limit of the two-way fixed effects DD estimator (assuming that T is fixed and N grows) in terms of potential outcomes and separates the estimand from the identifying assumptions:

$$\text{plim}_{N \rightarrow \infty} \hat{\beta}^{DD} = \beta^{DD} = VWATT + VWCT + \Delta ATT \quad (12)$$

The first term in (12) is the two-way fixed effects DD estimand, which I call the “variance-weighted average treatment effect on the treated” (VWATT):

$$\begin{aligned} VWATT \equiv & \sum_{k \neq U} \sigma_{kU} ATT_k(POST(k)) \\ & + \sum_{k \neq U} \sum_{\ell > k} \sigma_{k\ell} [\mu_{k\ell} ATT_k(MID(k, \ell)) + (1 - \mu_{k\ell}) ATT_\ell(POST(\ell))] \quad (12a) \end{aligned}$$

The σ terms correspond to the s terms in equation (7), but replace sample shares (n) with population shares (n_k^*).¹⁴ VWATT is always a positively weighted average of ATTs for the units *and* periods that act as treatment groups across the 2x2 DD’s that make up $\hat{\beta}^{DD}$. The weights come from the decomposition theorem and reflect group size and treatment variance.

The second term, which I call “variance-weighted common trends” (VWCT) generalizes common trends to a setting with timing variation:

$$\begin{aligned} VWCT \equiv & \sum_{k \neq U} \sigma_{kU} [\Delta Y_k^0(POST(k), PRE(k)) - \Delta Y_U^0(POST(k), PRE(k))] \\ & + \sum_{k \neq U} \sum_{\ell > k} \sigma_{k\ell} [\mu_{k\ell} \{\Delta Y_k^0(MID(k, \ell), PRE(k)) - \Delta Y_\ell^0(MID(k, \ell), PRE(k))\} \\ & + (1 - \mu_{k\ell}) \{\Delta Y_\ell^0(POST(\ell), MID(k, \ell)) - \Delta Y_k^0(POST(\ell), MID(k, \ell))\}] \quad (12b) \end{aligned}$$

¹⁴ Note that a DD estimator is not consistent if T gets large because the permanently turned on treatment dummy becomes collinear with the unit fixed effects ($\frac{X'X}{T}$ does not converge to a positive definite matrix). Asymptotics with respect to T require the time dimension to grow in both directions (see Perron 2006).

Like VWATT, VWCT is also an average of the difference in untreated potential outcome trends between pair of groups (and over different time periods) using the weights from the decomposition theorem. VWCT is new, it defines internal validity for the DD design with timing, and it is weaker than the more commonly assumed *equal* counterfactual trends across groups.

The last term in (12) equals a weighted sum of the *change* in treatment effects within each unit's post-period:

$$\Delta ATT \equiv \sum_{k \neq U} \sum_{\ell > k} \sigma_{k\ell} (1 - \mu_{k\ell}) [ATT_k(POST(\ell)) - ATT_k(MID(k, \ell))] \quad (12c)$$

Because already-treated groups sometimes act as controls, the 2x2 estimators in equation (11c) subtract average changes in their untreated outcomes *and* their treatment effects. Equation (12c) defines the bias that comes from estimating a single-coefficient DD model when treatment effects vary over time. Note that this does not mean that the DD research design is invalid. In this case other specifications, such as an event-study model (Jacobson, LaLonde, and Sullivan 1993) or “stacked DD” (Abraham and Sun 2018, Deshpande and Li 2017, Fadlon and Nielsen 2015), or other estimators such as reweighting strategies (Callaway and Sant'Anna 2018, de Chaisemartin and D'Haultfœuille 2018b) may be more appropriate.

Recent DD research comes to related conclusions about DD models with timing, but does not describe the full estimator as in equation (12). Abraham and Sun (2018), Borusyak and Jaravel (2017), and de Chaisemartin and D'Haultfœuille (2018b) begin by imposing pairwise common trends (VWCT=0), and then incorporating ΔATT into the DD estimand.¹⁵ The structure of the decomposition theorem, however, suggests that we should think of ΔATT as a source of bias

¹⁵ Abraham and Sun (2018, p 6) assume “parallel trends in baseline outcome”; Borusyak and Jaravel (2017, p 10) assume “no pre-trends”; and de Chaisemartin and D'Haultfœuille (2018b, p 6) assume “common trends” in counterfactual outcomes. Two of these papers analyze common specifications that I do not consider. de Chaisemartin and D'Haultfœuille (2018b) discuss designs where treatment evolves continuously within group-by-time cells and first-difference specifications. Abraham and Sun (2018) analyze the estimand for semi-parametric event-study models.

because it arises from the way equation (2) forms “the” control group. This distinction, made clear in equation (12), ensures an interpretable estimand (VWATT) and clearly defined identifying assumptions.¹⁶ This follows from at least two related precedents. de Chaisemartin and D’Haultfœuille (2018a, p. 5) prove identification of dose-response DD models under the assumption that “the average effect of going from 0 to d units of treatment among units with $D(0)=d$ is stable over time.” Treatment effect homogeneity ensures an estimand with no negative weights. Similarly, the monotonicity assumption in Imbens and Angrist (1994) ensures that the local average treatment effect does not have negative weights. In other words, negative weights are a failure of identification rather than a feature of the IV estimand.

A. Interpreting the DD Estimand

When the treatment effect is a constant, $ATT_k(W) = ATT$, $\Delta ATT = 0$, and $VWATT = ATT$. The rest of this section assumes that $VWCT=0$ and discusses how to interpret VWATT under different forms of treatment effect heterogeneity.

i. Effects that vary across units but not over time

If treatment effects are constant over time but vary across units, then $ATT_k(W) = ATT_k$ and we still have $\Delta ATT = 0$. In this case DD identifies:

$$VWATT = \sum_{k \neq U} ATT_k \left[\overbrace{\sigma_{kU} + \sum_{j=1}^{k-1} \sigma_{jk} (1 - \mu_{jk}) + \sum_{j=k+1}^K \sigma_{jk} \mu_{jk}}^{\equiv w_k^T} \right] \quad (13)$$

¹⁶ Equation (12) shows that the negative weighting pointed out elsewhere only occurs when treatment effects vary over time. The mapping between (12) and existing decompositions can be made explicit by rewriting all the $ATT_k(W)$ terms (see equation 9) in terms of group-time effects, which are the object of, for example, Theorem 1 in de Chaisemartin and D’Haultfœuille (2018b). In general, each group-time effect receives a total amount of weight that comes partly from strictly positive terms (equation 12b) and some from potentially negative terms (equation 12c). When treatment effects are constant, though, the negative weights in (12c) on the group-time effects in $ATT_k(MID(k, \ell))$ cancel with the positive weights on group-time effects in $ATT_k(POST(\ell))$. Then $\Delta ATT = 0$ and DD estimates VWATT, which has strictly positive weights.

VWATT weights together the group-specific ATTs not by sample shares, but by a function of sample shares and treatment variance. The weights in (13) equal the sum of the decomposition weights for all the terms in which group k acts as the treatment group, defined as w_k^T .

The parameter in (13) does not necessarily have a structural interpretation. In general, $w_k^T \neq n_k^*$, so the parameter does not equal the sample ATT.¹⁷ Neither are the weights proportional to the share of time each unit spends under treatment, so VWATT also does not equal the effect in the average treated period. The weights, specifically the central role of treatment variance, comes from the use of least squares. OLS combines 2x2 DD's efficiently by weighting them according to variances of the treatment dummy.¹⁸ VWATT lies along the bias/variance tradeoff: the weights deliver efficiency by potentially moving the point estimate away from, say, the sample ATT.

This tradeoff may not be worthwhile, particularly when VWATT differs strongly from a given parameter of interest, which occurs when treatment effect heterogeneity is correlated with treatment timing. Therefore, the processes that determine treatment timing are central to the interpretation of VWATT (see Besley and Case 2002). For example, a Roy model of selection on gains (where the number of units treated in each period is constrained) implies that treatment rolls out first to units with the largest effects. Site selection in experimental evaluations of training programs (Joseph Hotz, Imbens, and Mortimer 2005) and energy conservation programs (Allcott 2015) matches this pattern. In this case, regression DD underestimates the sample-weighted ATT if t_1^* is early enough (or there are a lot of “post” periods) so that $\bar{D}_1$ is very small and $\bar{D}_K \approx 0.5$, and overestimates it if t_1^* is late enough (or there are a lot of “pre” periods) so that $\bar{D}_1 \approx 0.5$ and

¹⁷ Abraham and Sun (2018), Borusyak and Jaravel (2017), Chernozhukov et al. (2013), de Chaisemartin and D'Haultfœuille (2018b), Gibbons, Serrato, and Urbancic (2018), Wooldridge (2005) all make a similar observation. The DD decomposition theorem, provides a new solution for the relevant weights.

¹⁸ This is exactly analogous to the result that two-stage least squares is the estimator that “efficiently combines alternative Wald estimates” (Angrist 1991).